The score plateau hides moving answers.

A question-level audit of all 19 clean GreekMMLU checkpoints, joined to exact source-token and document exposure from the immutable D0 schedule. The report also recovers the earlier Gemma, Krikri and Qwen comparison—with its historical-evidence limits made explicit.

> **From the accuracy peak to the endpoint, 1,137 formerly correct answers are lost and 821 new ones are gained. The net score falls—but 12.12% of all questions change correctness state.**

01 · TRAINING GEOMETRY AT THE 40B CHECKPOINT

Five clocks align near halfway—but none changes phase there.

The benchmark peak is placed against the executed learning-rate schedule, AdEMAMix's two full-run warmups, and exact HPLT/GlossAPI exposure. Curves are reconstructed from the executed optimizer configuration; GreekMMLU and exposure points are measured.

5.50×10^{-5}

learning rate at 40B · still peak

2.09

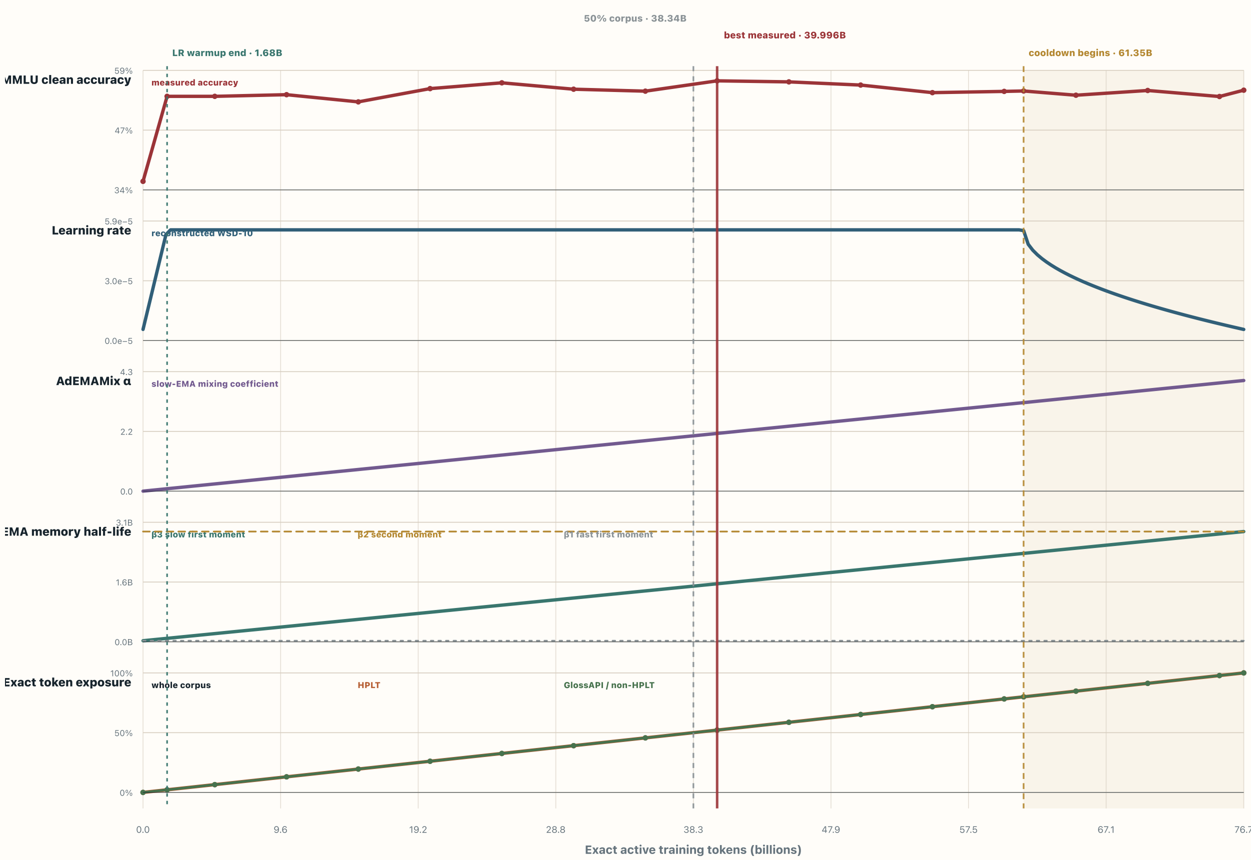
effective AdEMAMix α · target 4.0

0.998100

effective β3 · 364-update / 1.53B-token half-life

52.16%

complete corpus consumed · HPLT and GlossAPI match

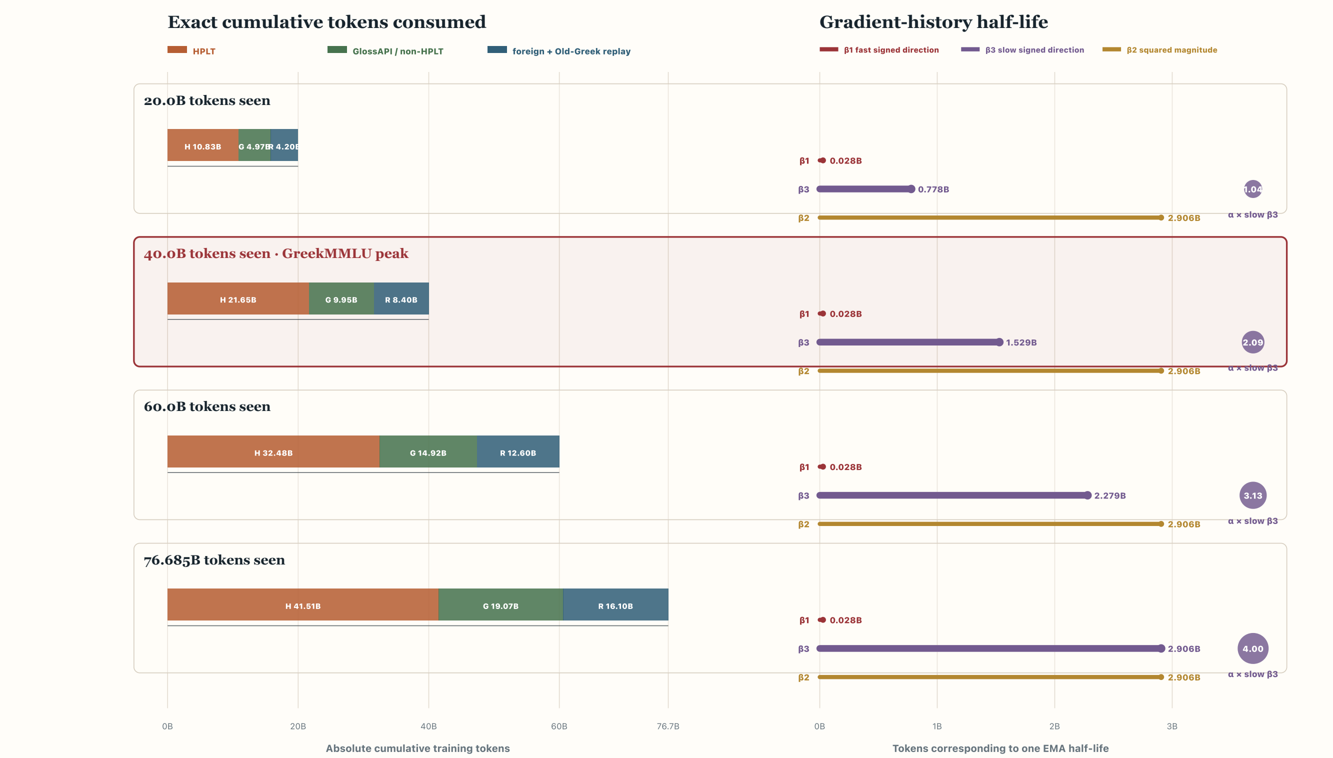


Solid points: measured benchmark and exposure evidence. Schedule curves: reconstructed from executed flags and the AdEMAMix half-life scheduler.

Red line: update 9,536 / 39.996B · grey line: exact corpus halfway.

How much data has been seen—and how far backward does each β look?

The left side shows cumulative source tokens. The right side shows each exponential moving average's half-life in training tokens. A half-life is not a hard cutoff: gradients older than the bar still contribute, but at progressively smaller weights. α changes the strength of the slow β3 branch; it does not change β3's memory length.



Left: exact cumulative tokens consumed from HPLT, GlossAPI/non-HPLT, and replay. Right: β1, β3 and β2 memory half-lives.

Purple α badge: coefficient applied to the slow β3 directional average.

No 40B switch

Learning rate is still at its peak, the cooldown has not started, and α and β3 are only partway through smooth full-run ramps. The coincidence is therefore an optimum along moving schedules—not a configured phase boundary.

Slow history gains influence

At the peak, α gives the slow first-moment branch a coefficient of about 2.09 while β3 gives it a roughly 364-update half-life. This is a mixing coefficient, not a claim that the slow branch contributes exactly twice the update magnitude.

All data clocks are halfway

HPLT and GlossAPI exposure track total corpus progress almost exactly. The graph therefore cannot identify HPLT’s 50% point as the cause of the benchmark peak.

Adaptation versus stability

Peak LR keeps individual steps plastic while the increasingly weighted slow EMA makes them depend on a longer history. That combination can keep source loss improving while benchmark decision boundaries continue to move.

02 · BENCHMARK TRAJECTORY

Accuracy peaks early; continuous loss does not settle identically.

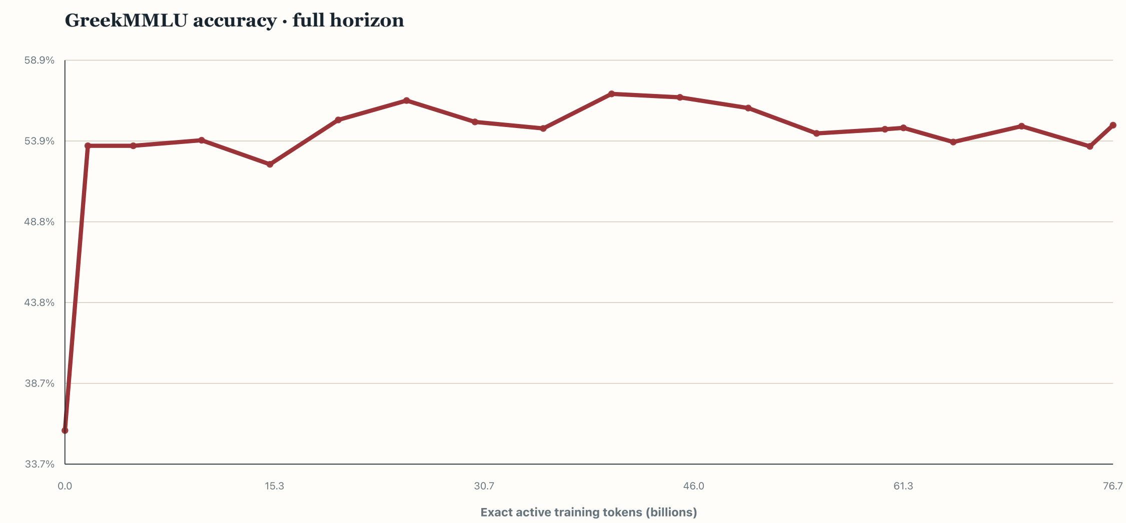
Every point uses the same decontaminated 16,159-question subset. Accuracy is a discrete argmax; choice NLL is the continuous probability assigned to the correct option relative to the alternatives.

56.81%
best clean accuracy · update 9,536

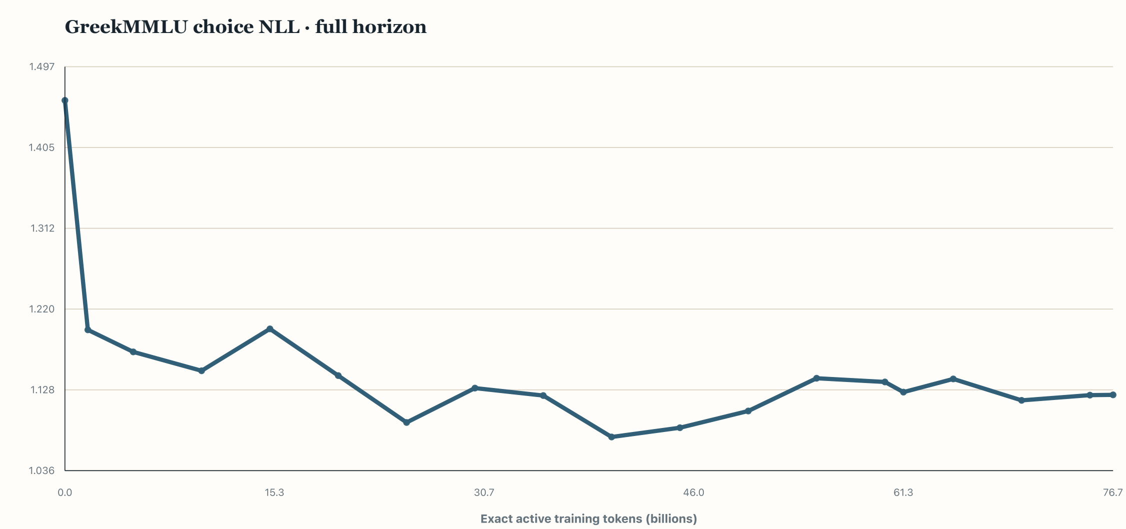
54.85%
final clean accuracy

15.22%
selected answer changed · peak → final

57.61%
correct at some, but not all, checkpoints



Clean zero-shot accuracy · higher is better. Exact active-token x-axis.



Clean multiple-choice NLL · lower is better. Same 19 checkpoints.

03 · QUESTION-LEVEL STATE CHANGES

A flat score is not a stable answer set.

At each transition, “newly correct” means wrong at the preceding checkpoint and correct now; “newly wrong” is the reverse. Choice flips count any change in the selected option, even when both choices are wrong.

Correctness crossings between adjacent checkpoints

Green: newly correct; red: newly wrong. Bars are percentages of the fixed 16,159 questions.

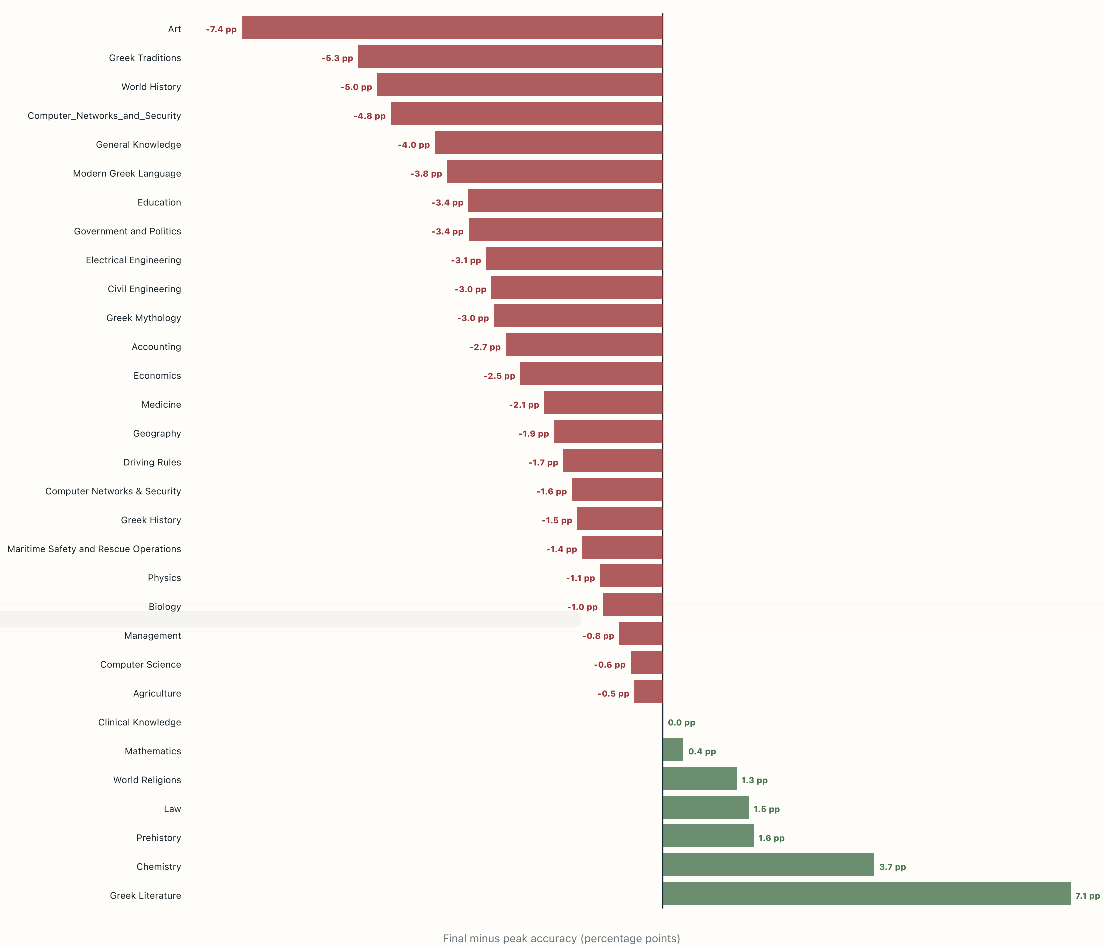
CHECKPOINT	ACTIVE TOKENS	ACCURACY	NEWLY CORRECT	NEWLY WRONG
0	0.000B	35.78%	—	—
400	1.678B	53.57%	4,376	1,502
1,192	5.000B	53.57%	807	807
2,384	9.999B	53.91%	670	614
3,576	14.999B	52.41%	488	731
4,768	19.998B	55.18%	1,144	696
5,960	24.998B	56.40%	773	577
7,152	29.997B	55.06%	544	760
8,344	34.997B	54.65%	661	727
9,536	39.996B	56.81%	939	590
10,728	44.996B	56.59%	593	628
11,920	49.995B	55.93%	807	915
13,112	54.995B	54.34%	783	1,039
14,304	59.994B	54.60%	662	620

CHECKPOINT	ACTIVE TOKENS	ACCURACY	NEWLY CORRECT	NEWLY WRONG
14,627	61.349B	54.69%	561	547
15,496	64.994B	53.80%	937	1,080

04 · AREAS OF KNOWLEDGE

Which subjects drift—and when?

The chart fixes the reference at update 9,536—the global accuracy peak—and asks which subject groups gained or lost by the final checkpoint. The first table then exposes every subject at every checkpoint, so local reversals are not hidden by the endpoint summary. Small subjects are shown but should be read with their sample size.



Final minus update-9,536 accuracy.

Positive is improvement; negative is regression.

Subject accuracy at every checkpoint

below that row's mean · near that row's mean · above that row's mean

Red border = row minimum · green border = row maximum

SUBJECT	N	U0 0.0B	U400 1.7B	U1,192 5.0B	U2,3 10.
Modern Greek Language	2,358	**34.52%**	52.21%	52.50%	54.2
Physics	1,828	**39.72%**	59.79%	58.92%	59.3
Driving Rules	1,265	**37.31%**	50.12%	50.83%	50.8
Mathematics	1,118	**28.80%**	55.55%	55.28%	**55.9**
Law	931	**28.14%**	36.09%	36.63%	38.0
Civil Engineering	767	**32.72%**	50.59%	50.33%	50.0
Greek History	669	**42.60%**	63.98%	64.72%	66.8
Management	660	**36.97%**	57.42%	56.52%	57.7
Art	651	**37.33%**	52.07%	50.38%	50.0
Clinical Knowledge	633	**38.07%**	57.35%	56.87%	57.1
Electrical Engineering	550	**35.27%**	49.45%	48.73%	46.0
Medicine	531	**31.83%**	48.59%	49.53%	47.8
Geography	421	**40.14%**	61.05%	64.85%	63.6
Agriculture	405	**30.12%**	51.11%	47.65%	47.4
Biology	382	**31.15%**	46.86%	47.64%	48.1
Greek Traditions	375	**37.87%**	61.07%	58.13%	60.0

Each cell is accuracy on the fixed clean subset for that subject. Color intensity is normalized within its own row, so it reveals when that subject performed relatively well or poorly; it does not compare difficulty across subjects. Hover for the row mean and exact deltas.

Subject endpoint summary

SUBJECT	N	INITIAL	BEST ACROSS RUN	BEST UPDA
Modern Greek Language	2,358	34.52%	56.83%	9,53
Physics	1,828	39.72%	63.46%	11,92
Driving Rules	1,265	37.31%	51.78%	5,96
Mathematics	1,118	28.80%	55.90%	2,38
Law	931	28.14%	38.67%	17,88
Civil Engineering	767	32.72%	55.02%	8,34
Greek History	669	42.60%	73.24%	9,53
Management	660	36.97%	60.91%	9,53
Art	651	37.33%	58.06%	5,96
Clinical Knowledge	633	38.07%	59.56%	7,15
Electrical Engineering	550	35.27%	51.82%	4,76
Medicine	531	31.83%	50.66%	10,72
Geography	421	40.14%	72.45%	5,96
Agriculture	405	30.12%	56.30%	10,72
Biology	382	31.15%	52.09%	5,96
Greek Traditions	375	37.87%	63.20%	9,53
Computer Science	358	42.74%	63.41%	5,96

Educational-level accuracy at every checkpoint

The same row-relative coloring and best/worst borders are applied here.

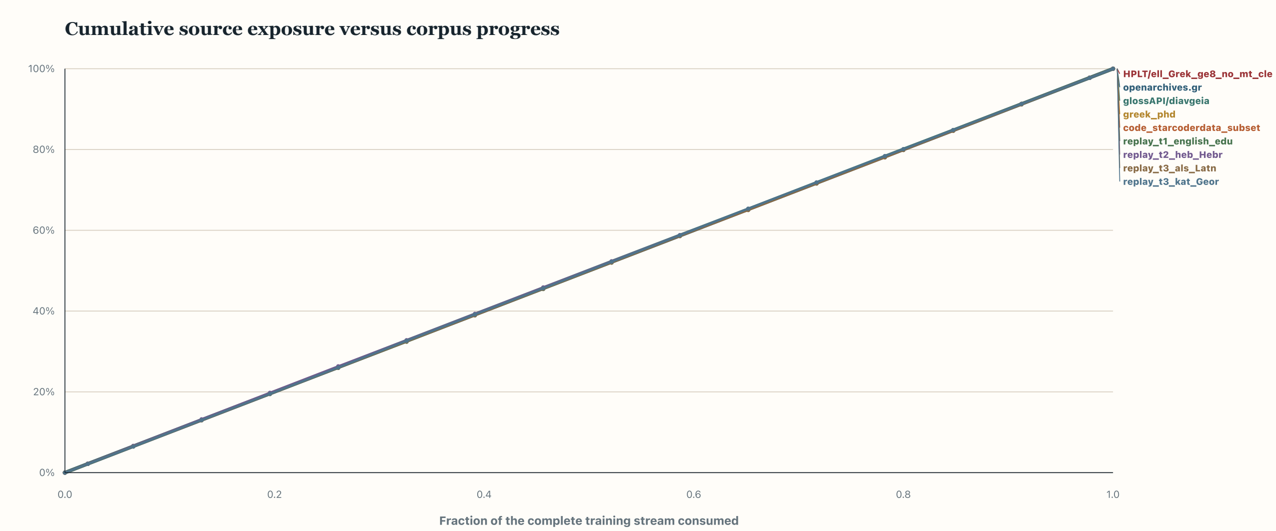
LEVEL	N	U0 0.0B	U400 1.7B	U1,192 5.0B	U2,384 10.0B	U3,576 15.0B
Professional	8,070	**34.60%**	51.54%	50.71%	50.90%	49.80%

LEVEL	N	U0 0.0B	U400 1.7B	U1,192 5.0B	U2,384 10.0B	U3,576 15.0B
Primary School	4,390	**36.86%**	58.95%	59.89%	60.36%	58.45%
Secondary School	1,523	**37.03%**	54.50%	54.69%	56.40%	52.66%
NA	1,265	**37.31%**	50.12%	50.83%	50.83%	51.46%
University	911	**36.88%**	48.85%	50.38%	49.62%	47.31%

05 · EXACT CORPUS EXPOSURE

What fraction of every source had actually been seen?

These are reconstructed loss-active tokens from the frozen packed sequence IDs—not nominal 79/20/1 quotas. "Touched" documents have at least one intersecting sequence consumed; "fully seen" documents have all intersecting sequences consumed.



Largest source families; diagonal means perfectly uniform exposure. Full source × checkpoint table follows.

SOURCE / CATEGORY	TOKENS	DOCUMENTS	0.0B	1.7B
HPLT/ell_Grek_ge8_no_mt_clean60	41.513B	46,535,439	0.00%	2.19%

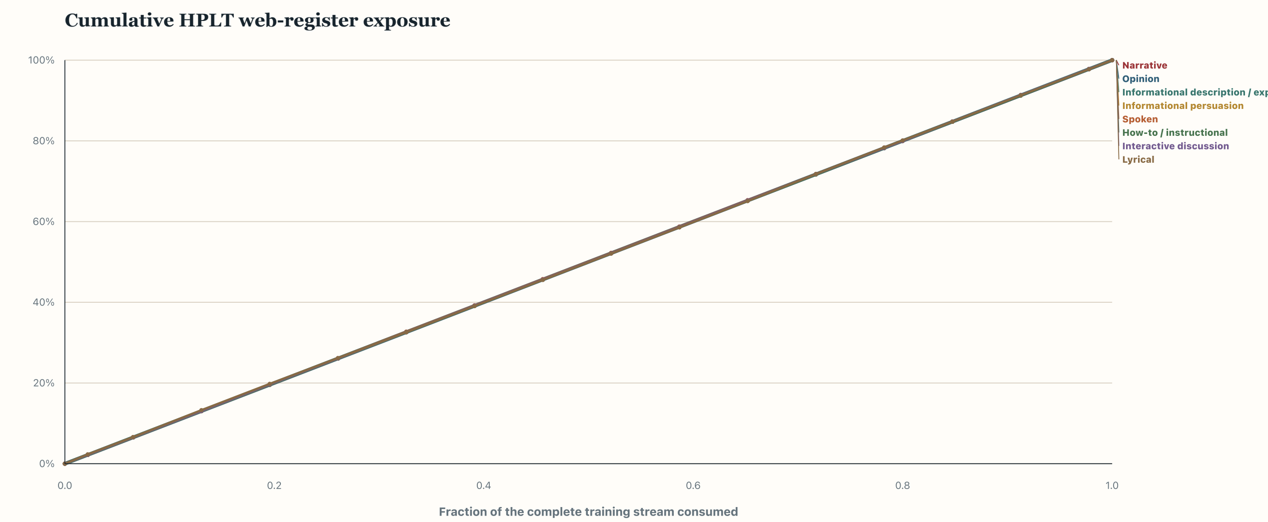
SOURCE / CATEGORY	TOKENS	DOCUMENTS	0.0B	1.7B
openarchives.gr	5.611B	119,731	0.00%	2.20%
glossAPI/diavgeia	4.604B	2,274,162	0.00%	2.19%
greek_phd	3.874B	31,620	0.00%	2.17%
code_starcoderdata_subset	2.686B	2,308,933	0.00%	2.17%
replay_t1_english_edu	1.422B	1,343,778	0.00%	2.19%
replay_t2_heb_Hebr	1.240B	975,435	0.00%	2.24%
replay_t3_als_Latn	1.181B	1,507,824	0.00%	2.18%
replay_t3_kat_Geor	1.157B	893,976	0.00%	2.16%
replay_t2_ron_Latn	1.092B	1,013,860	0.00%	2.21%
replay_t2_srp_Cyrl	1.012B	770,139	0.00%	2.20%
replay_t2_bul_Cyrl	1.001B	898,844	0.00%	2.14%
replay_t2_pes_Arab	0.980B	900,927	0.00%	2.19%
replay_t3_mkd_Cyrl	0.960B	1,324,628	0.00%	2.23%
greek_replay_apertus_original	0.767B	639,746	0.00%	2.19%
replay_t2_ukr_Cyrl	0.709B	737,335	0.00%	2.15%

Cell values are exact seen-token percentages. Hover a cell for touched- and fully-seen-document percentages.

06 · HPLT METADATA CATEGORIES

Which HPLT content types had been seen?

The available “type” field is HPLT’s preserved `register_level_1` text-register classifier, such as Narrative or Informational description/explanation. It describes the content, not necessarily the whole website, and it was not inferred after training.



Token exposure of HPLT register categories. Exact packed-document join.

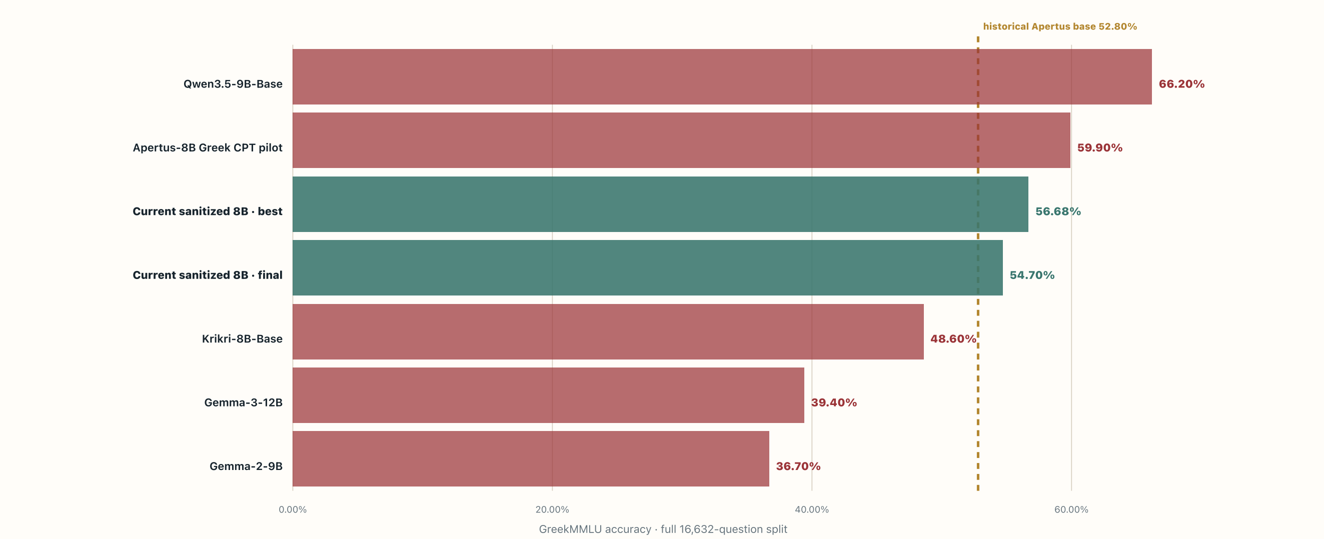
SOURCE / CATEGORY	TOKENS	DOCUMENTS	0.0B	1.7B	5.0B
Narrative	18.122B	27,281,821	0.00%	2.19%	6.51%
Opinion	8.961B	6,624,103	0.00%	2.18%	6.51%
Informational description / explanation	7.304B	4,148,536	0.00%	2.20%	6.55%
Informational persuasion	4.714B	6,574,002	0.00%	2.17%	6.51%
Spoken	0.966B	598,242	0.00%	2.21%	6.56%
How-to / instructional	0.769B	843,492	0.00%	2.20%	6.56%
Interactive discussion	0.564B	388,385	0.00%	2.22%	6.54%
Lyrical	0.113B	76,858	0.00%	2.33%	6.61%

07 · HISTORICAL MODEL CONTEXT

Gemma, Krikri, Qwen—and our earlier pilot.

The peer values were recovered from the preserved June 19 plot after the raw CSCS evaluation payloads had been pruned. They used the full 16,632-question split. The current run is receiptbacked and shown separately; the two evidence classes are context, not a controlled leaderboard.

RECOVERED PLOT EVIDENCE



Historical plot values plus current full-split best/final for orientation.

Public peers may contain GreekMMLU training overlap.

MODEL	ACCURACY	EVIDENCE CLASS	
Qwen3.5-9B-Base	66.20%	historical rendered plot	contamination-sensi
Apertus-8B Greek CPT pilot	59.90%	historical rendered plot	13.5B-token pilot; beta2=be
Current sanitized 8B · best	56.68%	current receipt	
Current sanitized 8B · final	54.70%	current receipt	
Krikri-8B-Base	48.60%	historical rendered plot	
Gemma-3-12B	39.40%	historical rendered plot	12B a
Gemma-2-9B	36.70%	historical rendered plot	

08 · INTERPRETATION

What the combined evidence says.

Plateau ≠ convergence

The aggregate accuracy wanders around a plateau while hundreds of individual questions cross the decision boundary between adjacent checkpoints.

Drift has subject structure

Most subject groups are below their update-9,536 value at the endpoint, but a minority continue improving. The endpoint is a different capability mix, not simply a uniformly worse copy.

D0 exposure is auditable

The stationary D0 schedule is empirically uniform: among every source of at least 1B tokens, the largest departure from overall corpus progress is only 0.25 percentage points. Across HPLT register categories, the maximum is 0.24 points.

No causal attribution from one run

The join can test whether a category had been seen; it cannot prove that a local exposure increment caused a subject score change. That requires controlled schedules or source-order seeds.

09 · REPRODUCIBILITY

Evidence boundaries and receipts.

/Users/foivoskarounos-zamparloukos/Projects/train-apertus-with-glossapi/subprojects/07_full_8b_cpt/presentations/data/full8_checkpoint_drift_20260811/greekmmlu_answer_drift.json

SHA-256 57375dd88a5381b43b377f193a70140c4982a06d73f3498b1bbde17b3af75c18 · 159,570 bytes

exposure

/Users/foivoskarounos-zamparloukos/Projects/train-apertus-with-glossapi/subprojects/07_full_8b_cpt/presentations/data/full8_checkpoint_drift_20260811/checkpoint_source_exposure.json

SHA-256 372f12a00c06490cbfb315cc6cc87a74435ee9dee9920609422b5bffe4f2173f · 684,714 bytes

peer

/Users/foivoskarounos-zamparloukos/Projects/train-apertus-with-glossapi/subprojects/07_full_8b_cpt/presentations/data/greekmmlu_peer_models_recovered_20260619.json

SHA-256 53999757bf75b1d34b488d55a486effe9606e0cfa8f5a6257b46d6b1f94a7f91 · 2,089 bytes

full results

/Users/foivoskarounos-zamparloukos/Projects/train-apertus-with-glossapi/subprojects/07_full_8b_cpt/presentations/FULL8_SANITIZED_CPT_FINAL_RESULTS_20260811.data.json

SHA-256 1706b3d85fdc485a44bb11c12db7388dedf535f704c9219104329d158bf3decd · 3,745,030 bytes

executed optimizer log

/Users/foivoskarounos-zamparloukos/Projects/train-apertus-with-glossapi/.codex_tmp/full8_current_comparison/new/logs/segment_4_training.log.pre_final_snapshot

SHA-256 ced2c9e237e969e2c3c51cf0895b6076e4cc103156eaf56fc42af6b355752d03 · 1,208,397 bytes

answer drift analyzer

`/Users/foivoskarounos-zamparloukos/Projects/train-apertus-with-glossapi/subprojects/07_full_8b_cpt/analysis/analyze_greekmmlu_answer_drift.py`

SHA-256 de424755dfa0aa51a547e0d7a9c6be8c112601a58d41d3a61a75bcdf8a3721f5 · 10,919 bytes

source exposure analyzer

`/Users/foivoskarounos-zamparloukos/Projects/train-apertus-with-glossapi/subprojects/07_full_8b_cpt/analysis/analyze_checkpoint_source_exposure.py`

SHA-256 d1f679af51b2d1680a9c163fcecc4b569128793d6040a34e09285af5f8f5d115 · 29,816 bytes

ademamix schedule audit

`/Users/foivoskarounos-zamparloukos/Projects/train-apertus-with-glossapi/subprojects/03_apertus_extension_and_embedding_adaptation/_archive/2026-05-24_2B_bakeoff_review/AUDIT_FINDINGS.md`

SHA-256 b98527d1070c8bc6e50565a1e032d89bd94328cb1c2b3b506247a83c1a8ad13c · 17,967 bytes

Exact schedule join

49,621,491 retained Modern-Greek documents joined across 431 frozen parquet files.

exact loss-active source-document tokens inside sequences whose D0 schedule position precedes the checkpoint boundary

Historical peer limit

The historical CSCS peer-model run directories and logs were found, but their payloads had been pruned. Values are therefore transcribed from the preserved rendered plot and are not receipt-recomputable.

Historical full-split context only. The current full-8B trajectory uses the decontaminated 16,159-item subset and float32 scoring, so it must not be ranked directly against these recovered full-split values without caveats.